



Immunological Aspects of the GIT — MCQ Practice

Microbiology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which is the most abundant antibody class in the intestinal lumen?

EASY

- A. Secretory IgA
- B. IgG
- C. IgM
- D. IgE
- E. IgD

2. Which cell secretes α -defensins and lysozyme in the intestinal epithelium?

EASY

- A. Goblet cell
- B. Paneth cell
- C. M cell
- D. Tuft cell
- E. Enteroendocrine cell

3. Which cell type transports antigen across the epithelium?

EASY

- A. Enterocyte
- B. Goblet cell
- C. Paneth cell
- D. M cell
- E. Plasma cell

4. Which Peyer's patch feature is correct?

EASY

- A. They are found in the stomach
- B. Density increases toward the terminal ileum
- C. They are absent in the small intestine
- D. They contain only T cells
- E. They secrete mucus

5. A Crohn's patient is found to have Paneth cell abnormalities. What does this mean?

MEDIUM

- A. Paneth cell defects are associated with Crohn's disease
- B. They cause celiac disease
- C. They cause lactose intolerance
- D. They cause food allergy
- E. They have no clinical relevance

6. Which cytokines dominate the homeostatic gut environment?

MEDIUM

- A. IL-10 and TGF- β
- B. IL-4 and IL-5
- C. IL-1 and IL-6
- D. IL-25 and IL-33
- E. TNF and IFN- γ

7. ILC3 and Th17 cells help maintain barrier health by producing:

MEDIUM

- A. IgE
- B. Histamine
- C. IL-22, which stimulates epithelial AMPs
- D. Mucin directly
- E. TGF- β only

8. Where is secretory IgA produced and how does it reach the lumen?

MEDIUM

- A. Enterocytes; secreted directly
- B. Peyer's patch T cells; diffusion
- C. Goblet cells; mucus
- D. Plasma cells in the lamina propria; transported through epithelial cells
- E. Mast cells; degranulation

9. Which is a function of IgA in the gut?

MEDIUM

- A. Activates complement
- B. Triggers mast-cell degranulation
- C. Causes Th2 skewing
- D. Kills bacteria directly
- E. Blocks uptake of antigen into the systemic immune system

10. Which epithelial cell senses microbial metabolites and releases hormones such as serotonin?

MEDIUM

- A. Goblet cell
- B. Enteroendocrine cell
- C. Paneth cell
- D. M cell
- E. Tuft cell

11. A patient has a food allergy. Which mechanism is typical of an IgE-mediated reaction?

MEDIUM

- A. Th1 activation only
- B. Th2 cytokines → class switching → IgE → mast-cell degranulation
- C. Complement lysis
- D. Bile acid signalling
- E. Loss of tight junctions only

12. Which statement about skin-prick testing in food allergy is correct?

MEDIUM

- A. It is the gold standard
- B. It is reliable in intolerance
- C. It replaces clinical history
- D. It has limited predictive value; elimination diets are used
- E. It diagnoses lactose intolerance

13. Which are components of the gut-associated lymphoid tissue?

MEDIUM

- A. Gallbladder and spleen
- B. Pancreas and liver
- C. Peyer's patches, appendix and isolated lymphoid follicles
- D. Esophagus and stomach
- E. Thymus and bone marrow

14. A patient with disordered gut motility has bacteria detected in the bloodstream. Which sequence best explains this?

MEDIUM

- A. Bacterial overgrowth → increased permeability → translocation
- B. Excess IgA → bacterial killing
- C. Increased acid → bacteremia
- D. Lactase deficiency → infection
- E. Tuft cell activation → sepsis

15. A patient with diarrhea and vomiting after milk, with hives and wheeze. Which best describes the mechanism?

HARD

- A. Lactase deficiency
- B. Fructose malabsorption
- C. Histamine intolerance only
- D. Celiac disease
- E. An immune-mediated food allergy

16. A patient with impaired tight junctions is at risk of:

HARD

- A. Achlorhydria
- B. Lactose intolerance
- C. Peyer's patch loss
- D. Pancreatic AMP excess
- E. Leaky gut with systemic inflammation, e.g. IBD

17. A child fails to develop tolerance and has excess Th2 activity. Which cell type would you expect to be increased in the gut wall?

HARD

- A. Neutrophils
- B. Paneth cells
- C. Eosinophils
- D. Treg cells
- E. M cells

18. Why are ILFs considered important sites of IgA production?

HARD

- A. They store IgE
- B. They secrete mucins
- C. They lack B cells
- D. They are a primary source of IgA-secreting B cells and change with the microbiome
- E. They only exist in the stomach

19. A patient has a bile-acid signalling imbalance. Which disease link does the lecture make?

HARD

- A. Celiac disease only
- B. Food allergy
- C. Lactose intolerance
- D. Gastric cancer
- E. Crohn's disease and ulcerative colitis

20. Which pair correctly matches Th subsets to their role in the lamina propria?

HARD

- A. Th1: allergens; Th2: intracellular pathogens
- B. Th1: parasites; Th2: bacteria
- C. Th1: intracellular pathogens; Th2: allergens and parasites
- D. Th17: IgE production
- E. Treg: neutrophil recruitment

✓ Answer key

Q	Answer	Difficulty	Why
1	A	EASY	sIgA is the most abundant antibody in the gut lumen.
2	B	EASY	Paneth cells secrete antimicrobial peptides.
3	D	EASY	M cells are specialised for antigen transcytosis.
4	B	EASY	Peyer's patches lie along the small intestine, denser toward the terminal ileum.
5	A	MEDIUM	Paneth cell abnormalities are associated with many diseases including Crohn's.
6	A	MEDIUM	Anti-inflammatory IL-10 and TGF- β maintain tolerance.
7	C	MEDIUM	IL-22 stimulates epithelial cells to produce protective AMPs; these cells can also drive inflammation.
8	D	MEDIUM	IgA is made by lamina-propria plasma cells and transported across the epithelium.
9	E	MEDIUM	sIgA neutralises, blocks adherence and blocks antigen uptake.
10	B	MEDIUM	Enteroendocrine cells sense microbial metabolites and secrete hormones that regulate gut physiology.
11	B	MEDIUM	Th2 cytokines drive IgE and mast-cell degranulation.
12	D	MEDIUM	Diagnosis is difficult; clinical suspicion and elimination diets are used.
13	C	MEDIUM	GALT includes Peyer's patches, the appendix and ILFs.
14	A	MEDIUM	Disordered motility leads to SIBO, permeability and bacterial translocation.
15	E	HARD	Involvement of skin and respiratory tract indicates an immune-mediated allergy.
16	E	HARD	Disturbed tight junctions allow luminal toxins, bacteria and antigens into tissues.
17	C	HARD	Th2 responses recruit eosinophils; eosinophils contribute to parasitic and allergic responses.
18	D	HARD	ILFs vary with microbiota and cytokines and are a primary source of IgA-secreting B cells.
19	E	HARD	Bile acids modulate macrophages and DCs; imbalance is implicated in IBD.
20	C	HARD	Th1 = cell-mediated (intracellular pathogens); Th2 = antibody-mediated (allergens, parasites).